



TOR VERGATA
UNIVERSITÀ DEGLI STUDI DI ROMA



Distributed Simulation Systems

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Distributed Simulation System

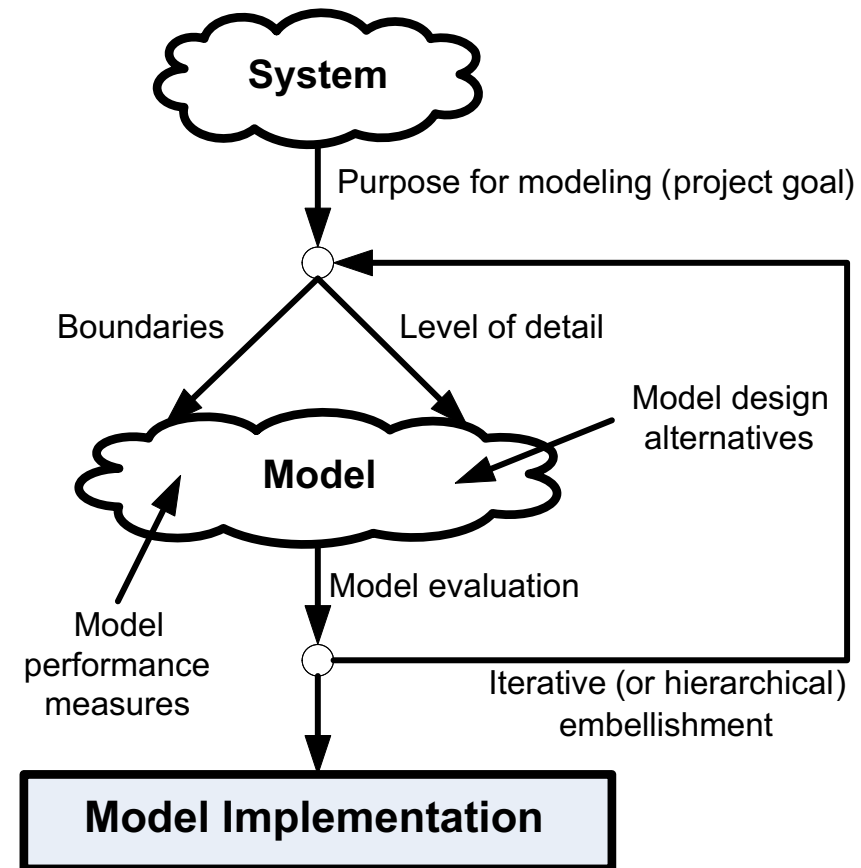
- Course objectives
 - To introduce methods for Internet-based distributed simulation
 - To illustrate relevant simulation paradigms and technologies
- Course topics
 - Local (Computer-based) simulation paradigm
 - Distributed simulation paradigm
 - Locally and geographically distributed simulation
 - Middleware for distributed simulation: the High-Level Architecture (HLA)
- Project assignment
 - Design and implementation of an HLA-based distributed simulation system
- Teaching material and course announcements
 - Provided on the MS Teams platform

Simulation vs DS vs Internet-based DS systems

- **Simulating** a system consists in
 1. defining a (simulation) **model** that represents the system of interest from a given point of view/level of abstraction,
 2. **execute the model** to carry out experiments on the model to deduct behaviour/properties/performance of the system
- A **distributed simulation (DS)** systems provide a flexible and low-cost solution to deal with the *computational, interoperability* and *scalability* properties required by modern simulation scenarios, without the need of using costly and complex dedicated platforms. Generally, DS systems have to deal with heterogeneity
- An **Internet-based DS** system is
 - a computer simulation system implemented as a software that is
 - executed on top of an Internet-based execution platform

Computer simulation

- A simulation effort carried out by use of a *computer-based platform*
- Consists of:
 - **designing** a model of a system
 - **implementing** the model as an executable software
 - **executing** the software onto a computer-based platform
 - **analyzing** the execution output



Modeling and Simulation (M&S) Lifecycle

Simulation vs. execution platform

- *Local (or Sequential, or Centralized) Simulation (LS)*
 - the simulation software is executed on top of a **single processor host** (i.e., a single processor computer)
- *Parallel Simulation (PS)*
 - the simulation software is executed on top of a **multi-processor host** (i.e., a computer with multiple processors connected through an “inter-processor” network)
- *Distributed simulation (DS)*
 - the simulation software is executed on top of **multiple hosts** (i.e., a set of computers connected through a network infrastructure of LAN, MAN or WAN type)
 - in the Internet-based DS case, the computer network makes use of Internet protocols

Example multi-processor host

Intel iPSC/2



Caratteristiche iPSC/2

processor

Intel 80386/80387 (16MHz)

number of
processors

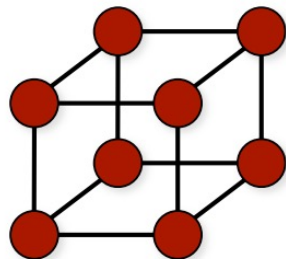
16

intern real
memory

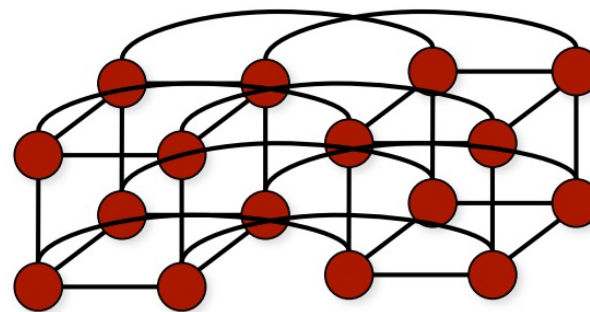
64 Mbyte: 4 Mbyte per processor

topology

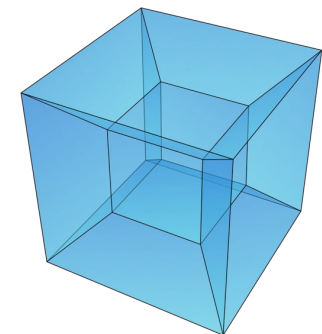
4-dimensional hypercube



3-dimensional hypercube
(n=8)



3-dimensional hypercube
(n=16)



Caratteristiche iPSC/2

interconnection	Direct Connect Module (DCM) with 8 full-duplex serial channels (2.8Mbyte/s) per processor. Chaneel 0 to 6 are used to connect to the neighbor processors.
OS	Node Executive (NX/2), a restricted OS with support for multitasking and communication.

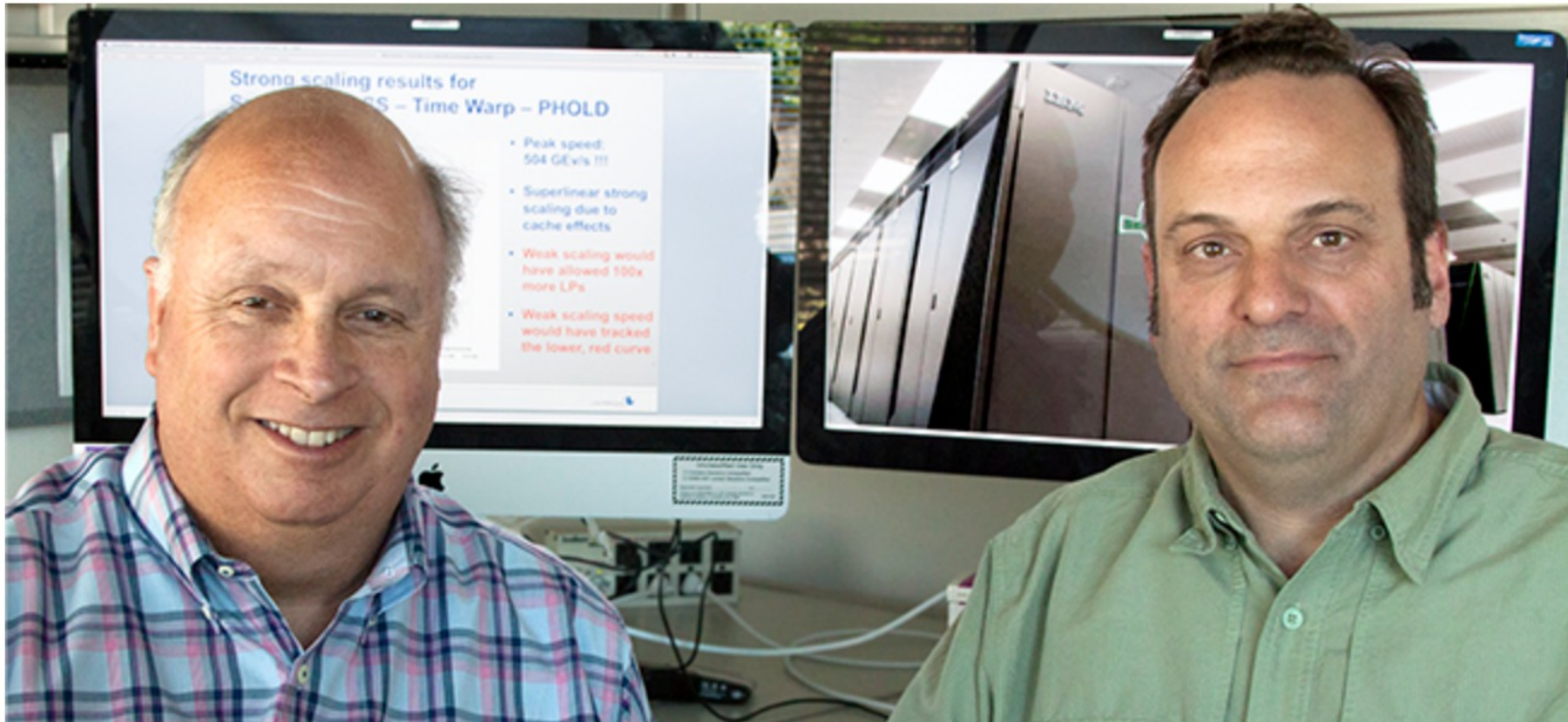
date of purchase 1987

used until 1993

Price 7.826.774 BEF

Simulation speed record (Apr. 13)

Lawrence Livermore and Rensselaer Polytechnic Institute scientists set a new *simulation speed record* on the Sequoia supercomputer



Lawrence Livermore scientists, from left, David Jefferson and Peter Barnes.

Simulation speed record: 504 billion events per second!

A step back to...the nature of Simulation

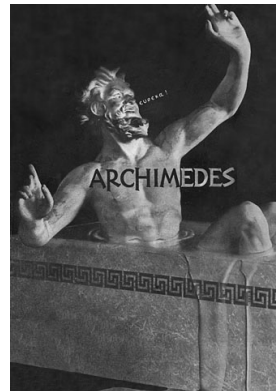
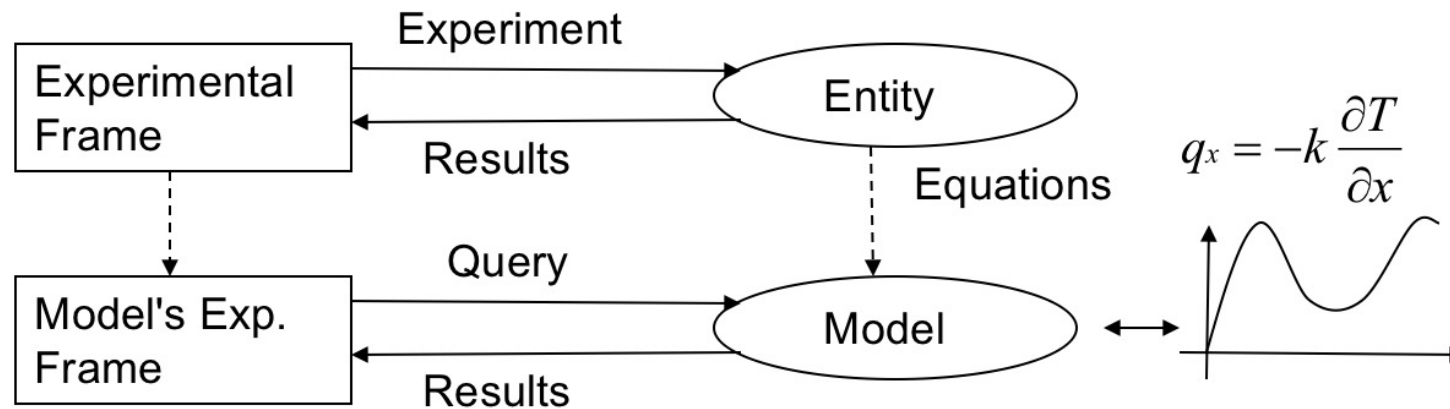
- To simulate, in the scientific connotation, is:
 “To reproduce the behavior of a given system, process or phenomenon by use of a model”
- To simulate, in simple terms, implies to *imitate* or *emulate*
- Simulation is the act of executing, experimenting with or exercising a model for a specific objective (intended use) such as *problem solving, training, acquisition, entertainment, research and education*

The nature of Simulation

- What's being simulated is the **physical system** (or *system under investigation*), which can be:
 - an **actual** (i.e., already existing) system
 - a **hypothetical** (i.e., to be realized) system, which is being analyzed and designed
- To study system, often make assumptions/approximations, both logical and mathematical, about how it works (*abstraction*)
- These assumptions form a **model** of the system, i.e., an abstract representation of the system (*principle of substitutability*)

The nature of Simulation

- If model structure is simple enough, could use mathematical methods to get exact information on questions of interest — *analytical solution*



$$\frac{\text{density of object}}{\text{density of fluid}} = \frac{\text{weight}}{\text{weight of displaced fluid}}$$